



Otsu Method & Segmentation

CSE0613409

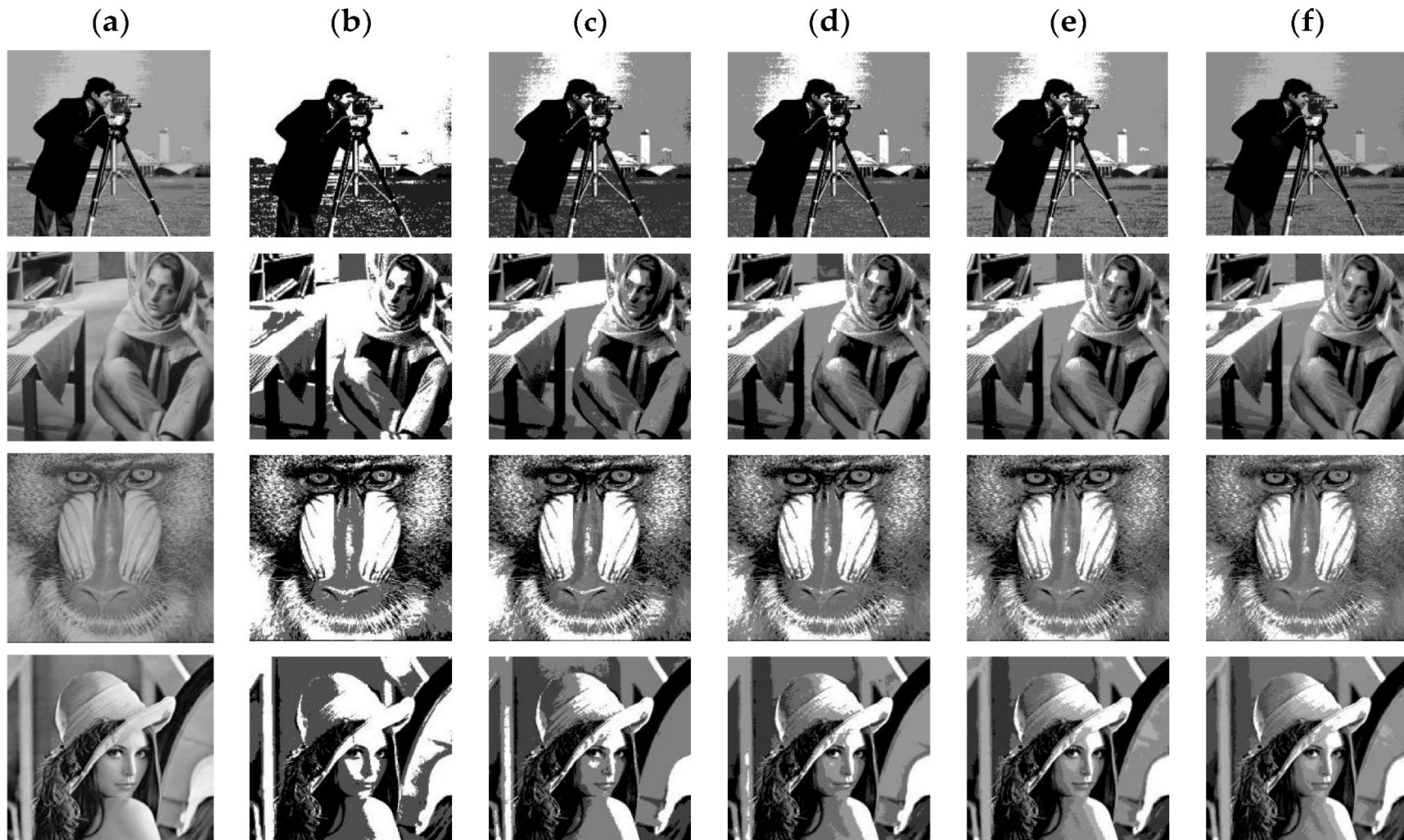
Machine Vision and Motion Analysis

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(MMF)

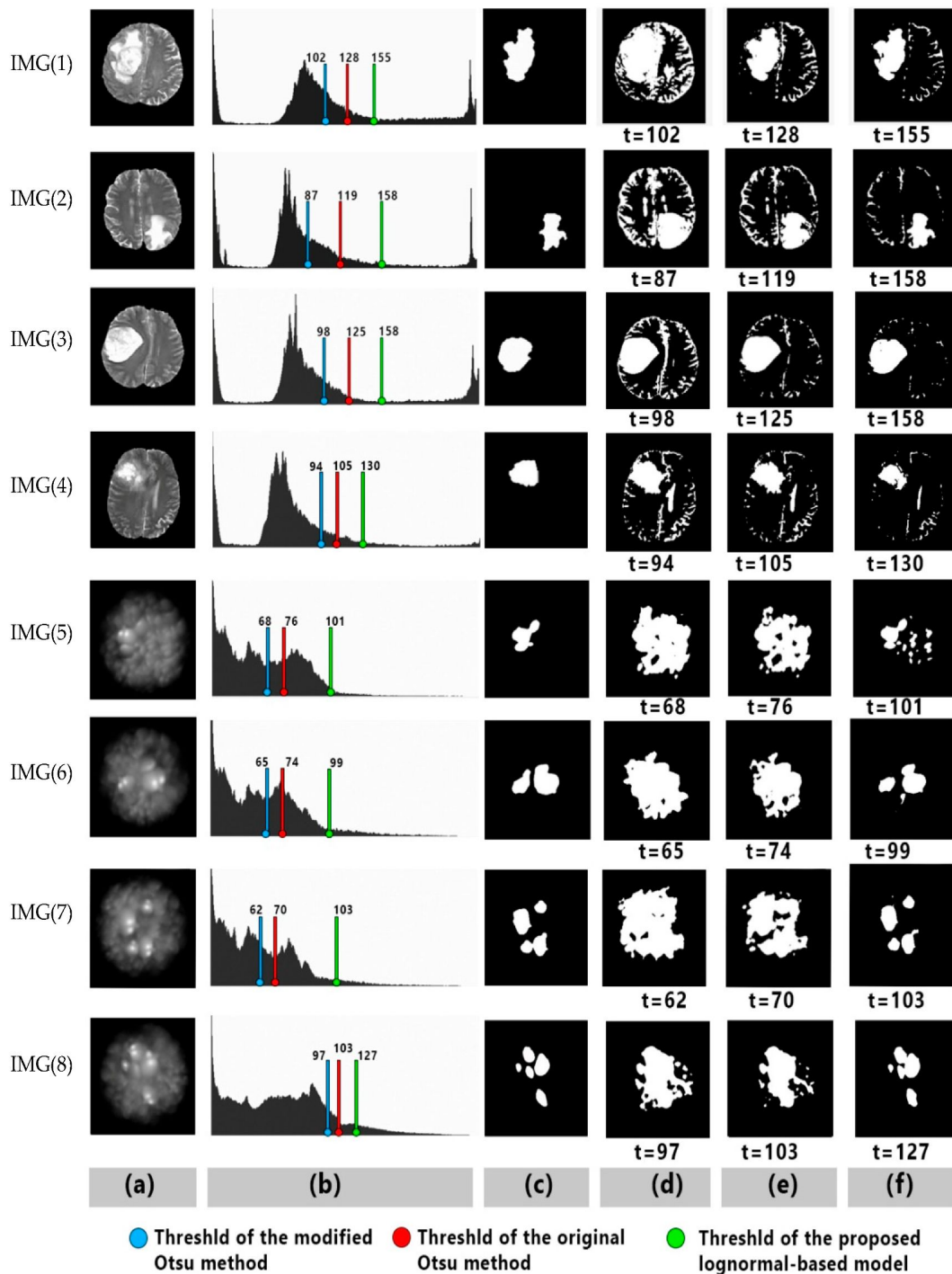
Department of Computer Science and Engineering

Why we need Otsu's method

Otsu's method is needed to automatically find the **best threshold value** for separating an image into a foreground and background without manual trial and error



Examples of Otsu's method



CT No.	Original image	Histogram of Original Image	Otsu's - optimal Segmented Image	Histogram of Otsu Segmented Image
CT-1				
CT-2				
CT-3				
CT-4				
CT-5				
CT-6				
CT-7				
CT-8				
CT-9				
CT-10				

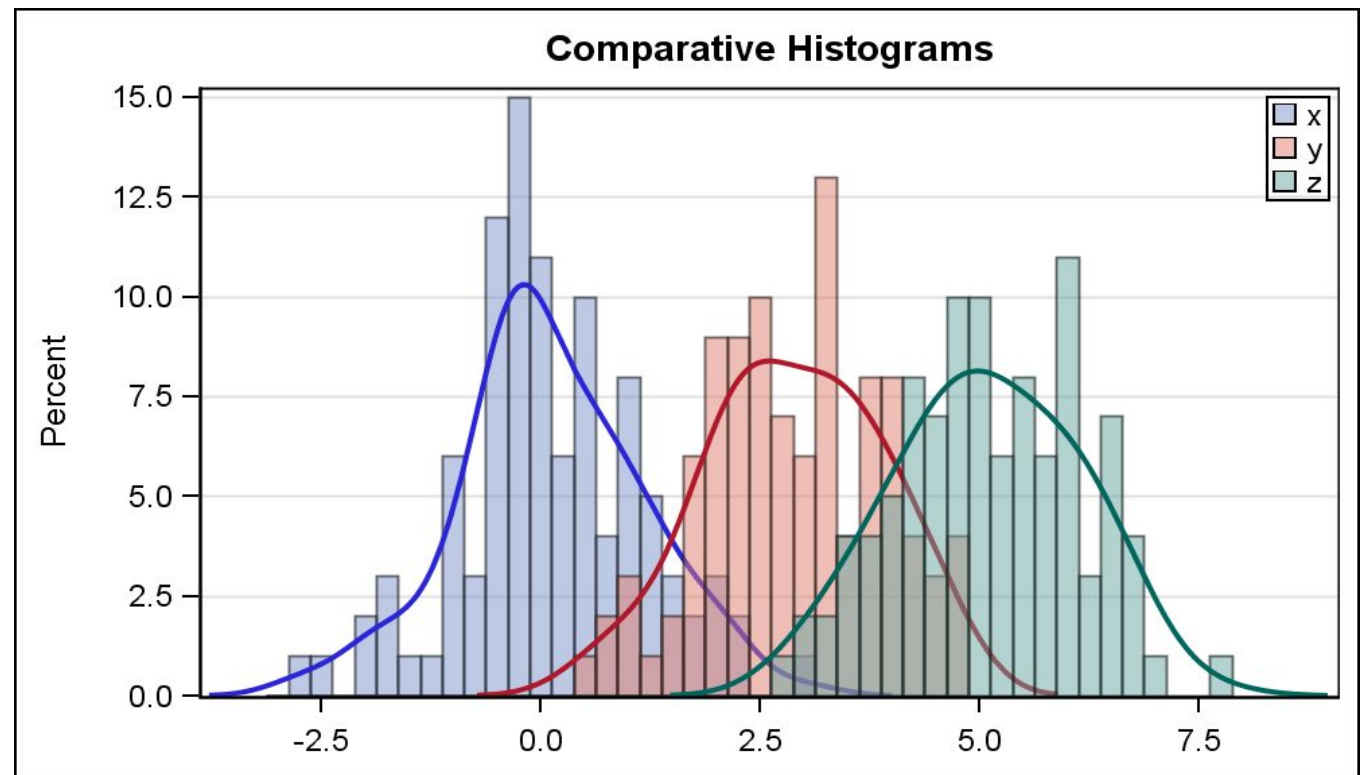
Histogram

What is a Histogram?

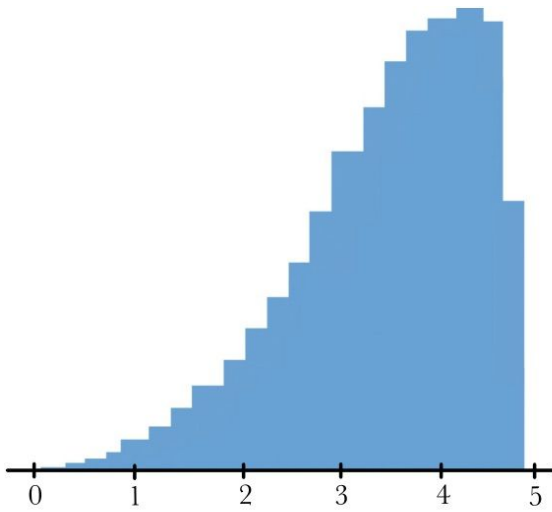
A histogram is a type of graph that shows how often numerical data falls into specific ranges.

For a grayscale image:

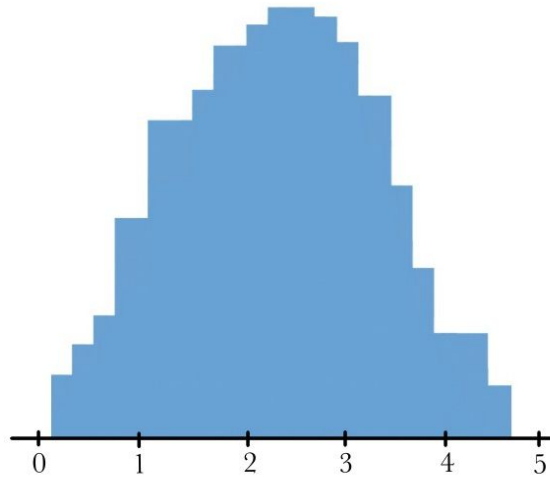
- **X-axis** → **Pixel intensity**
- **Y-axis** → **Number of pixels / Frequency**
- **0** → **Black**
- **255** → **White**



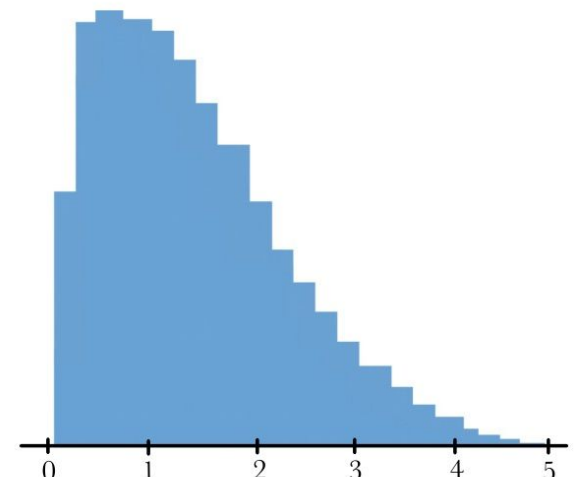
Histogram Shapes



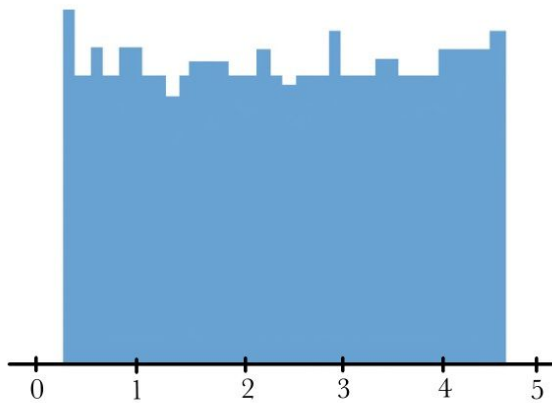
skew left



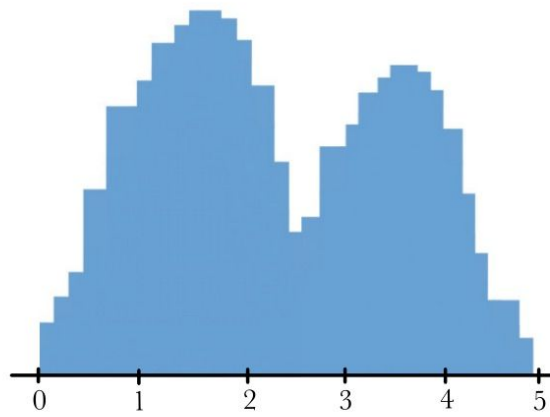
symmetric, unimodal



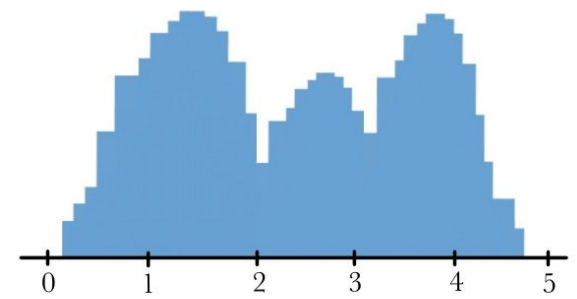
skew right



uniform



bimodal

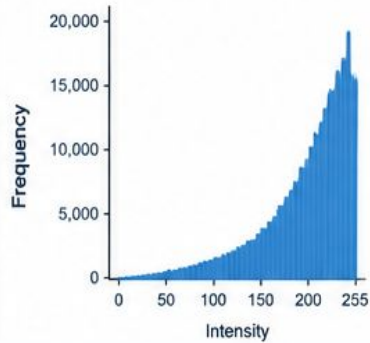


multimodal

Histogram Shapes

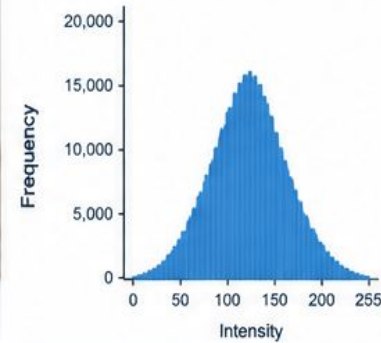
The shape of a histogram tells us how pixel intensities are distributed in an image.

Skew Left



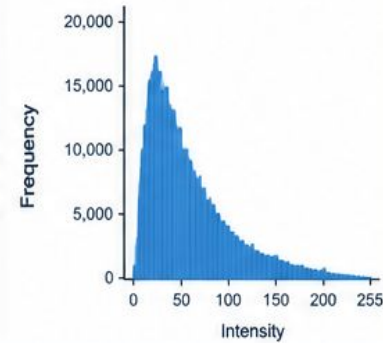
Most pixels are bright, with a longer tail toward darker intensities (left).

Symmetric / Unimodal



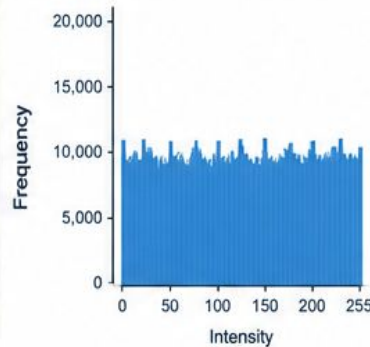
Pixels are balanced around midtones, with one central peak.

Skew Right



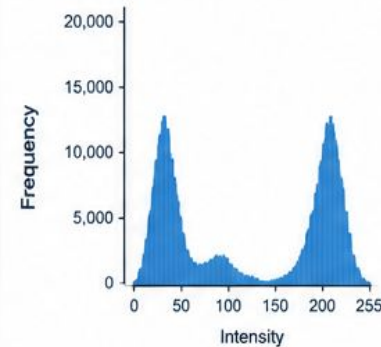
Most pixels are dark, with a longer tail toward brighter intensities (right).

Uniform



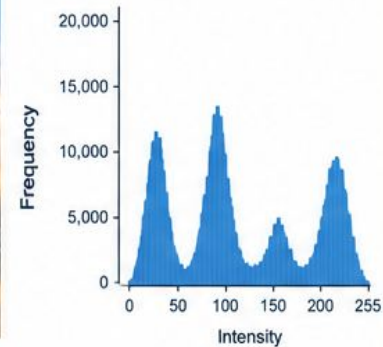
Pixel intensities are spread fairly evenly across the range.

Bimodal



Two distinct groups of intensities (e.g., dark object and bright background).

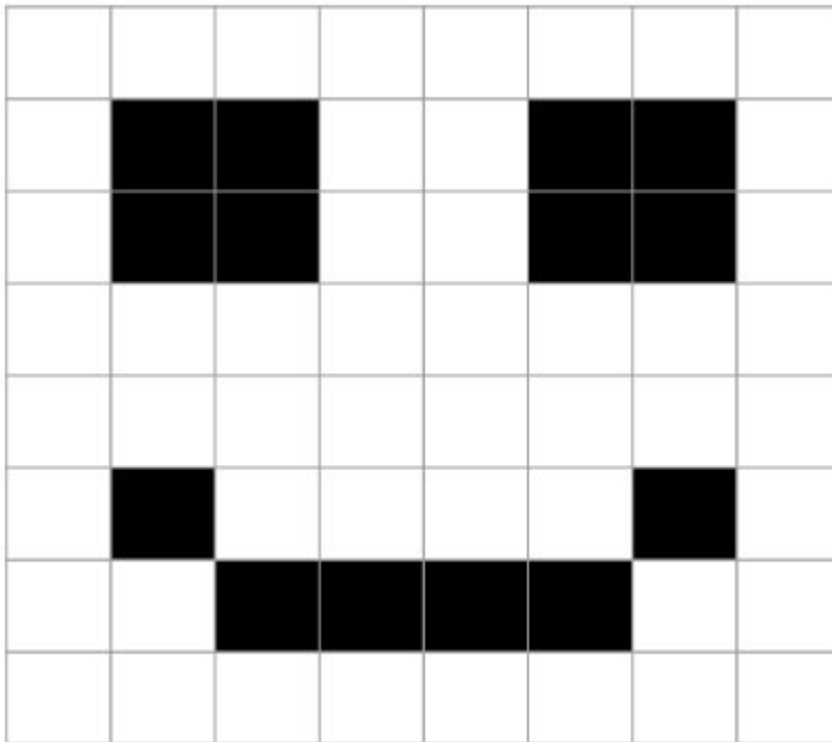
Multimodal



Three or more distinct intensity groups (multiple peaks).

Monochrome / Binary Image

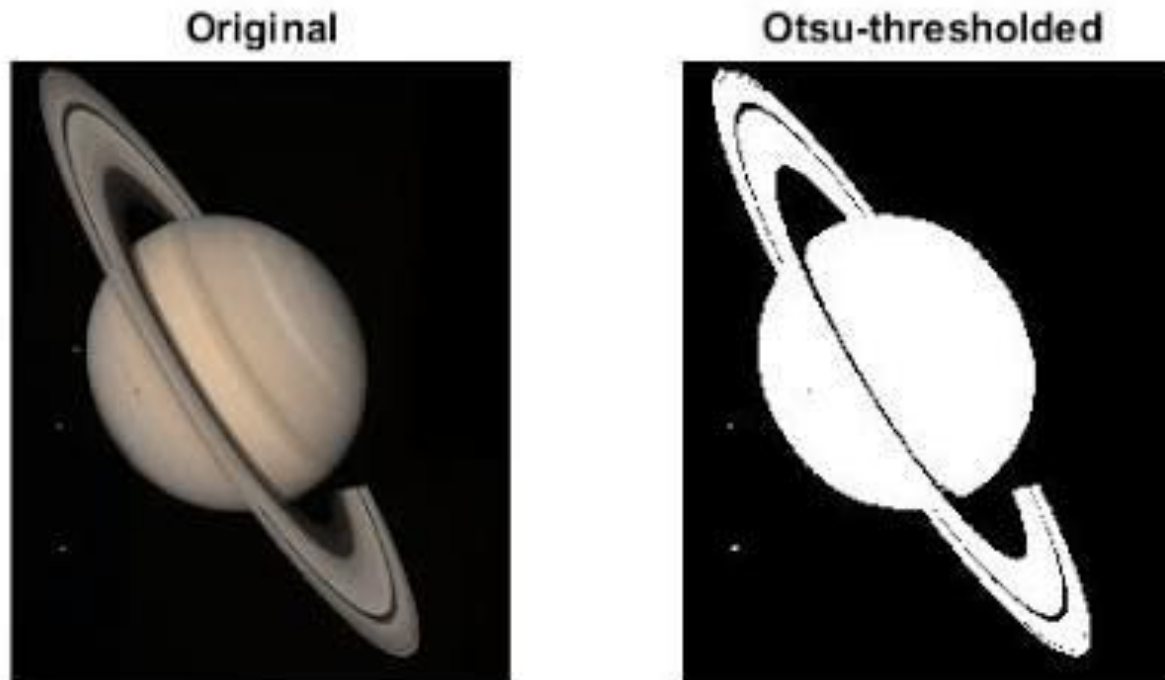
Monochrome means an image represented using shades of a single color.



0	0	0	0	0	0	0	0
0	1	1	0	0	1	1	0
0	1	1	0	0	1	1	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	1	0	0	0	0	1	0
0	0	1	1	1	1	0	0
0	0	0	0	0	0	0	0

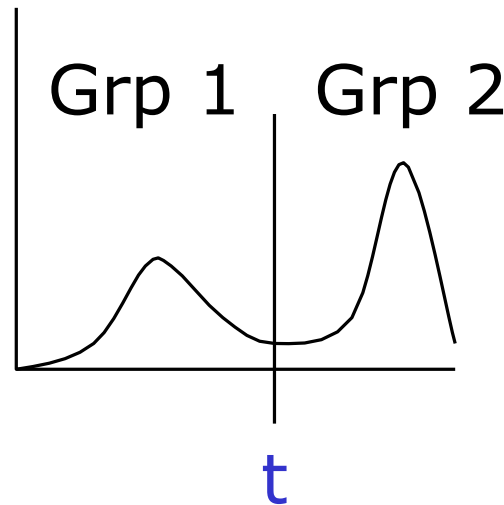
Otsu Thresholding

- Otsu thresholding is an automatic method for finding a threshold that separates an image into two classes.
- Or, Converting a grayscale image to Monochrome/binary image.
- **Otsu's method**, named after its inventor *Nobuyuki Otsu*, is one of many **binarization** algorithms.



Cont.

- This method *involves iterating* through all the possible threshold values and calculating a **measure of spread** for the pixel levels each side of the threshold, *i.e. the pixels that either fall in foreground or background*
- The aim is to find the threshold value where the sum of foreground and background **spreads is at its minimum.**



Assumption: the histogram is bimodal

Spread

Spread

Spread essentially means how far the pixel values are from their group's average.



Mathematically, this is represented using **variance**.

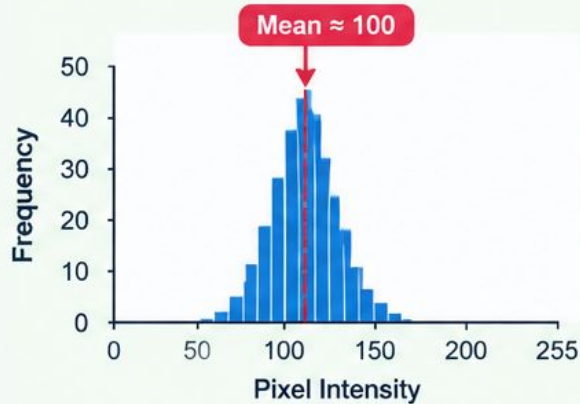
- If values inside a group are very close together:

Variance is small.

- If values are widely scattered:

Variance is large.

Small Variance (Tightly Clustered)



The pixel values are close to the mean.

→ **Small spread**

(low variance)



More uniform appearance

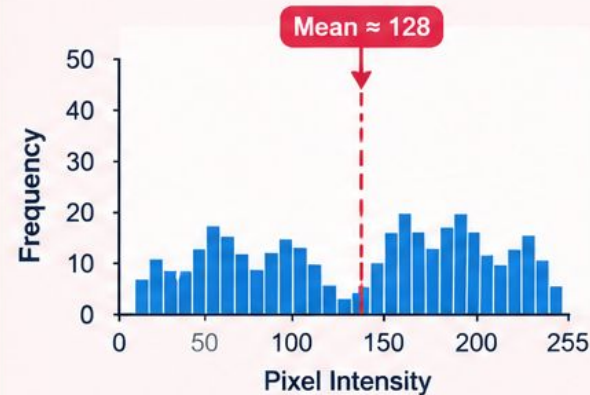
Example values

98 100 102 101 99 100

Mean = 100

Variance ≈ 2.3 (small)

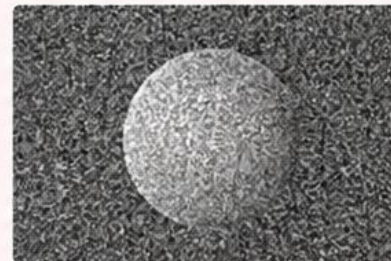
Large Variance (Widely Scattered)



The pixel values are far from the mean.

→ **Large spread**

(high variance)



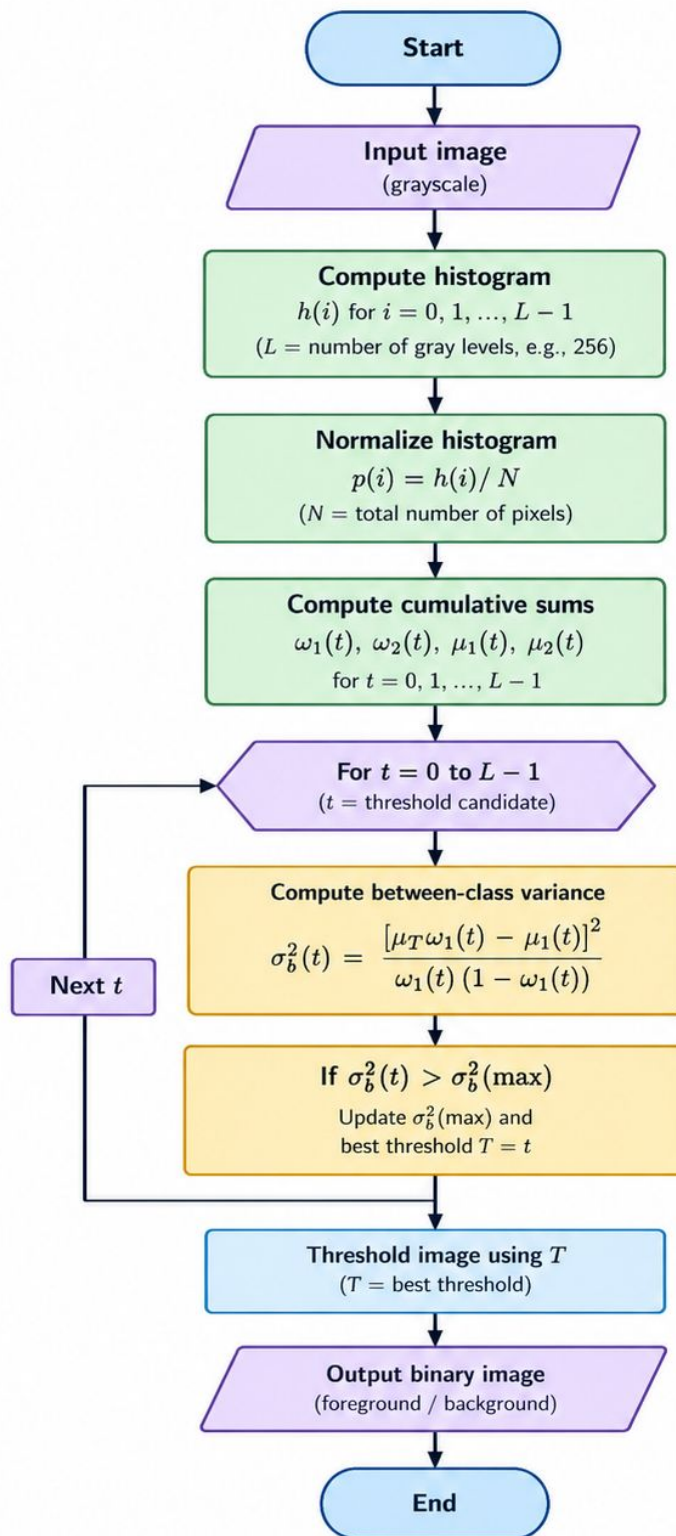
More varied appearance

Example values

30 70 120 160 210 240

Mean = 128

Variance $\approx 4,900$ (large)



Key Formulas

Normalized histogram:

$$p(i) = \frac{h(i)}{N}, \quad \sum_{i=0}^{L-1} p(i) = 1$$

Class probabilities:

$$\omega_1(t) = \sum_{i=0}^t p(i) \quad \omega_2(t) = \sum_{i=t+1}^{L-1} p(i)$$

Class means:

$$\mu_1(t) = \frac{\sum_{i=0}^t ip(i)}{\omega_1(t)} \quad \mu_2(t) = \frac{\sum_{i=t+1}^{L-1} ip(i)}{\omega_2(t)}$$

Total mean (constant):

$$\mu_T = \sum_{i=0}^{L-1} ip(i)$$

Between-class variance:

$$\sigma_b^2(t) = \frac{[\mu_T \omega_1(t) - \mu_1(t)]^2}{\omega_1(t) (1 - \omega_1(t))}$$

Variables

i	: gray level (0 to $L - 1$)
$h(i)$	: histogram count
$p(i)$	: normalized histogram
N	: total number of pixels
L	: number of gray levels (e.g., 256)
t	: threshold candidate
$\omega_1(t)$	: background probability
$\omega_2(t)$	: foreground probability
$\mu_1(t)$	: background mean
$\mu_2(t)$	: foreground mean
μ_T	: total mean
$\sigma_b^2(t)$	: between-class variance
T	: optimal threshold

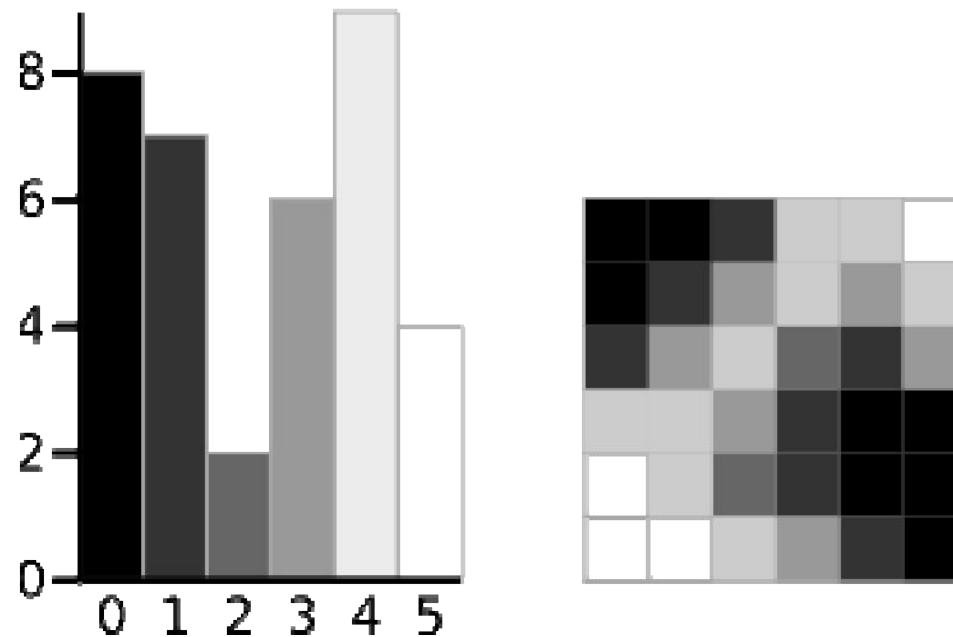
Result

$$T = \arg \max_t \sigma_b^2(t)$$

(threshold value)

Cont.

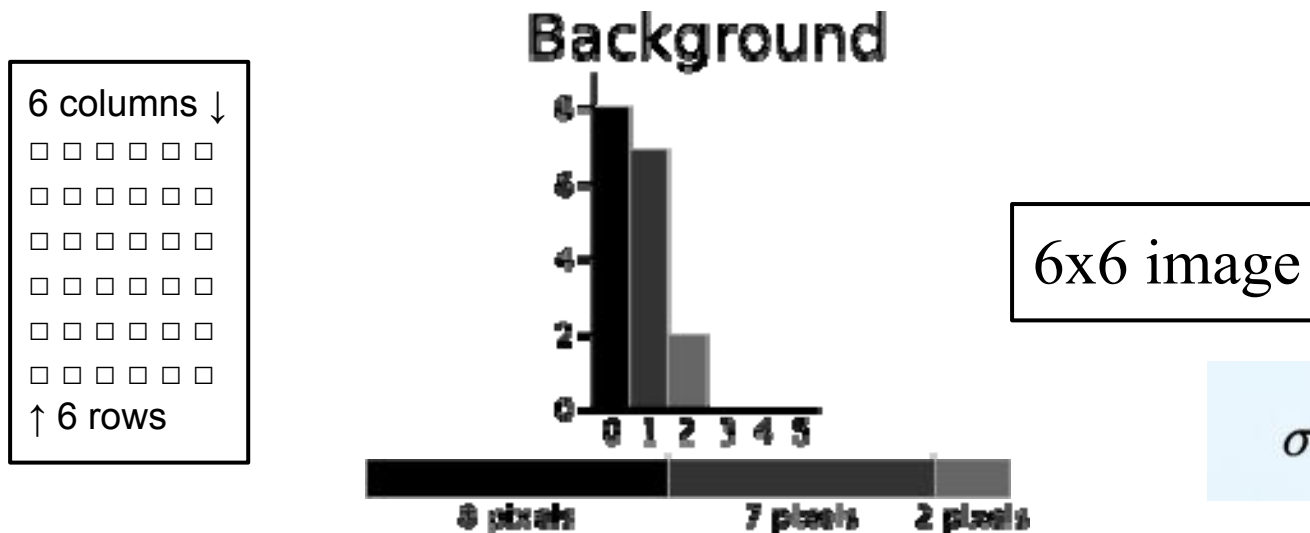
- Simple 6x6 image shown below and the histogram for the image is shown next
- To simplify the explanation, only 6 grayscale levels are used



A 6-level grayscale image and its histogram

Cont. (For Background)

- The calculations are bellow for finding the **foreground & background** variances (*the measure of spread*) for a single threshold



Weight $W_b = \frac{8 + 7 + 2}{36} = 0.4722$

Mean $\mu_b = \frac{(0 \times 8) + (1 \times 7) + (2 \times 2)}{17} = 0.6471$

Variance $\sigma_b^2 = \frac{((0 - 0.6471)^2 \times 8) + ((1 - 0.6471)^2 \times 7) + ((2 - 0.6471)^2 \times 2)}{17}$

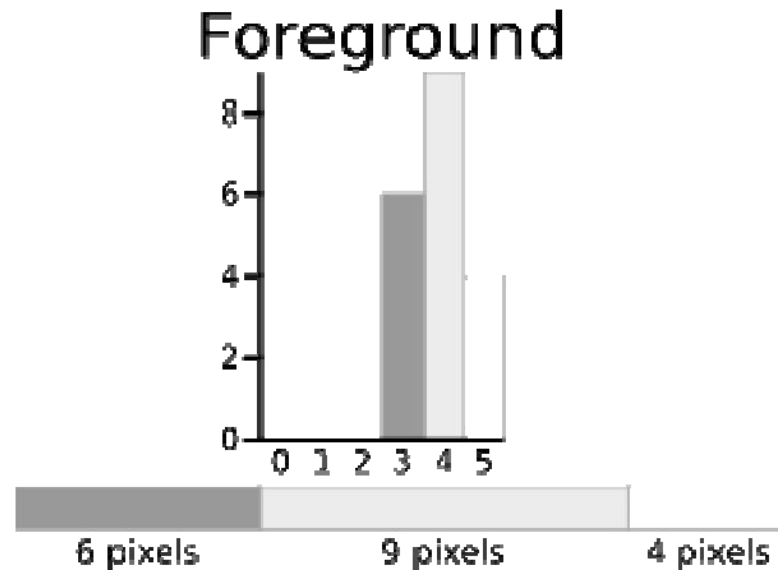
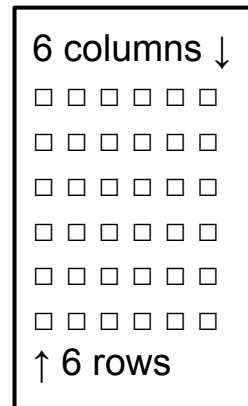
$$= \frac{(0.4187 \times 8) + (0.1246 \times 7) + (1.8304 \times 2)}{17}$$

$$= 0.4637$$

Variables

σ^2	= variance of the class
x_i	= intensity value of pixel i
μ	= mean intensity of the class
N	= number of pixels in the class

Cont. (For Foreground)



6x6 image

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$

Weight $W_f = \frac{6 + 9 + 4}{36} = 0.5278$

Mean $\mu_f = \frac{(3 \times 6) + (4 \times 9) + (5 \times 4)}{19} = 3.8947$

Variance $\sigma_f^2 = \frac{((3 - 3.8947)^2 \times 6) + ((4 - 3.8947)^2 \times 9) + ((5 - 3.8947)^2 \times 4)}{19}$

$$= \frac{(4.8033 \times 6) + (0.0997 \times 9) + (4.8864 \times 4)}{19}$$

$$= 0.5152$$

Variables

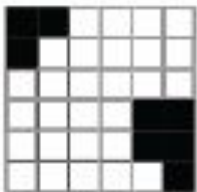
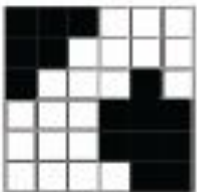
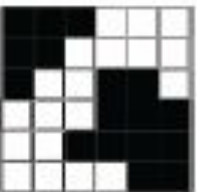
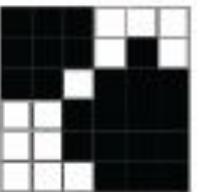
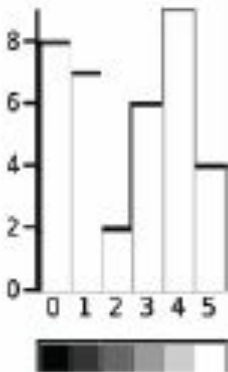
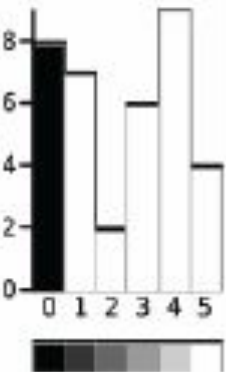
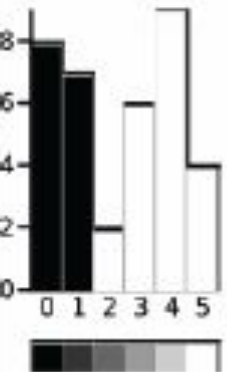
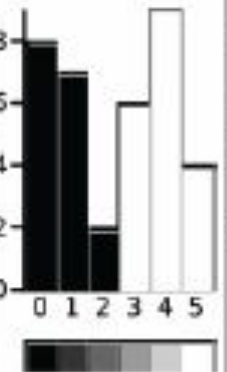
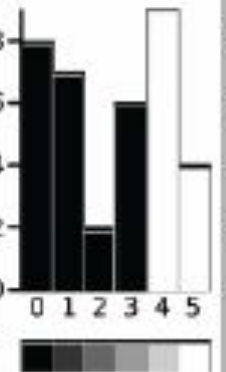
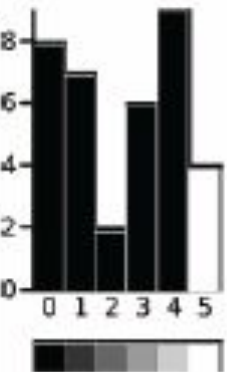
- σ^2 = variance of the class
- x_i = intensity value of pixel i
- μ = mean intensity of the class
- N = number of pixels in the class

Cont.

- The next step is to calculate the '*Within-Class Variance*' i.e. this is simply the sum of the two variances multiplied by their associated weights

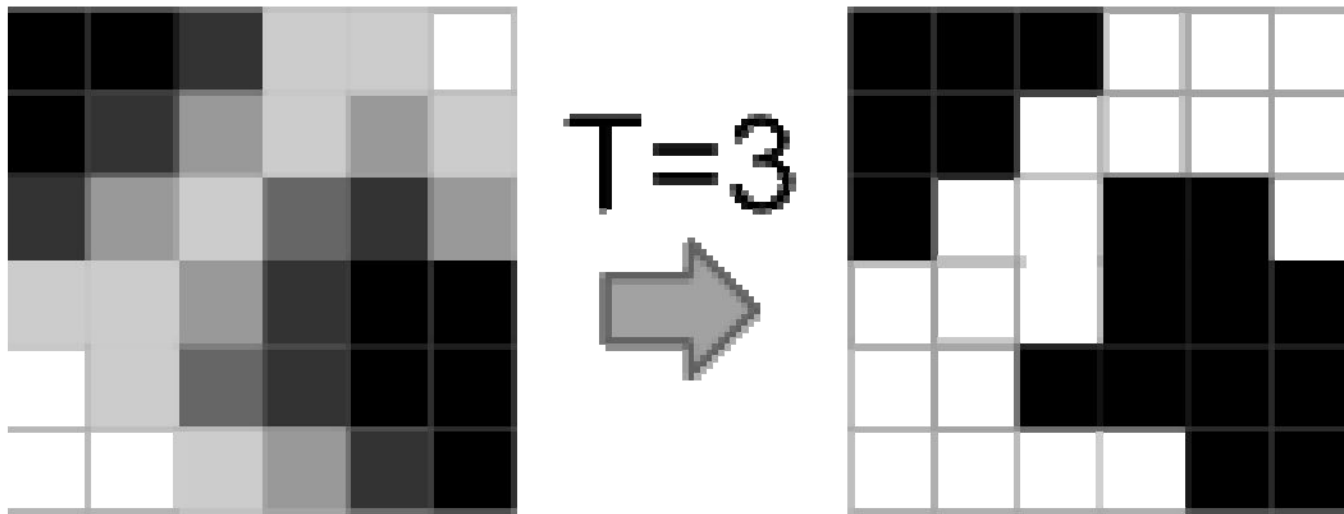
$$\begin{aligned}\text{Within Class Variance } \sigma_W^2 &= W_b \sigma_b^2 + W_f \sigma_f^2 = 0.4722 * 0.4637 + 0.5278 * 0.5152 \\ &= 0.4909\end{aligned}$$

Cont.

Threshold	T=0	T=1	T=2	T=3	T=4	T=5
						
						
Weight, Background	$W_b = 0$	$W_b = 0.222$	$W_b = 0.4167$	$W_b = 0.4722$	$W_b = 0.6389$	$W_b = 0.8889$
Mean, Background	$M_b = 0$	$M_b = 0$	$M_b = 0.4667$	$M_b = 0.6471$	$M_b = 1.2609$	$M_b = 2.0313$
Variance, Background	$\sigma_b^2 = 0$	$\sigma_b^2 = 0$	$\sigma_b^2 = 0.2489$	$\sigma_b^2 = 0.4637$	$\sigma_b^2 = 1.4102$	$\sigma_b^2 = 2.5303$
Weight, Foreground	$W_f = 1$	$W_f = 0.7778$	$W_f = 0.5833$	$W_f = 0.5278$	$W_f = 0.3611$	$W_f = 0.1111$
Mean, Foreground	$M_f = 2.3611$	$M_f = 3.0357$	$M_f = 3.7143$	$M_f = 3.8947$	$M_f = 4.3077$	$M_f = 5.000$
Variance, Foreground	$\sigma_f^2 = 3.1196$	$\sigma_f^2 = 1.9639$	$\sigma_f^2 = 0.7755$	$\sigma_f^2 = 0.5152$	$\sigma_f^2 = 0.2130$	$\sigma_f^2 = 0$
Within Class Variance	$\sigma_W^2 = 3.1196$	$\sigma_W^2 = 1.5268$	$\sigma_W^2 = 0.5561$	$\sigma_W^2 = 0.4909$	$\sigma_W^2 = 0.9779$	$\sigma_W^2 = 2.2491$

the threshold equal to 3, as well as being used for the example, also has the lowest sum of weighted variances

Conclusion



Result of Otsu method

Between class variance

Within Class Variance $\sigma_W^2 = W_b \sigma_b^2 + W_f \sigma_f^2$ (as seen above)

Between Class Variance $\sigma_B^2 = \sigma^2 - \sigma_W^2$
 $= W_b(\mu_b - \mu)^2 + W_f(\mu_f - \mu)^2$ (where $\mu = W_b \mu_b + W_f \mu_f$)
 $= W_b W_f (\mu_b - \mu_f)^2$

Threshold	T=0	T=1	T=2	T=3	T=4	T=5
Within Class Variance	$\sigma_W^2 = 3.1196$	$\sigma_W^2 = 1.5268$	$\sigma_W^2 = 0.5561$	$\sigma_W^2 = 0.4909$	$\sigma_W^2 = 0.9779$	$\sigma_W^2 = 2.2491$
Between Class Variance	$\sigma_B^2 = 0$	$\sigma_B^2 = 1.5928$	$\sigma_B^2 = 2.5635$	$\sigma_B^2 = 2.6287$	$\sigma_B^2 = 2.1417$	$\sigma_B^2 = 0.8705$

Thank You